

INFOMDW: Data Wrangling

Heterogeneous Data Integration

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Data integration

- Organizations use databases: mainframes, workstations, servers
 - Built from scratch each time... usually
 - Many models of the same object
- Need for information sharing across databases
 - Exploit advances in distributed computing and networking

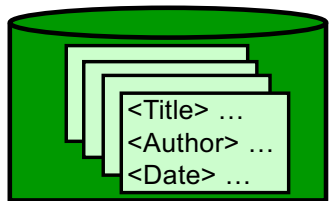


Data integration

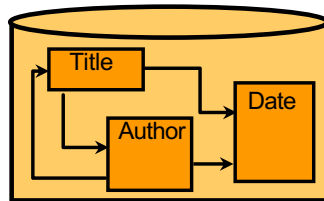
- Fact: Databases are managed by different persons
- Challenge 1: Provide uniform access to the users
- Challenge 2: Allow DB to interoperate but ...
 - Autonomous
 - Different OS
 - Different purposes
 - Different data modeling
 - Different data formats
 - Different access and communication protocols



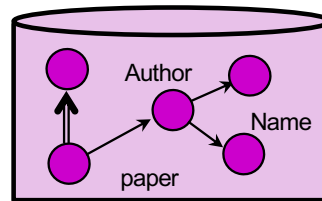
Information is everywhere



Web pages



RDBMS



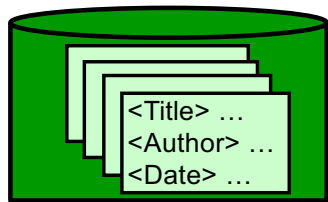
OODBMS

- Databases .. but not only
- Mail messages
- Web pages
- Spreadsheets
- Text documents
- Multimedia annotations
- XML Documents

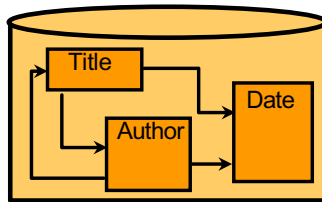


There are many needs

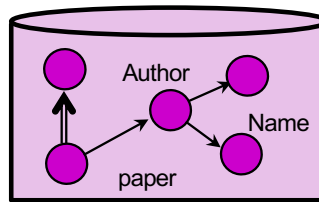
- Exchanged
- Replicated
- Migrated
- Integrated
- Adapted



Web pages



RDBMS



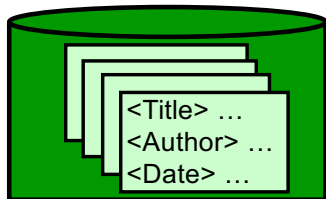
OODBMS



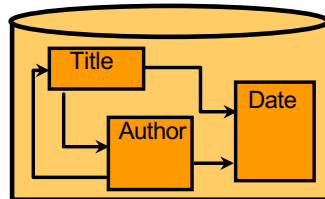
Data in different forms



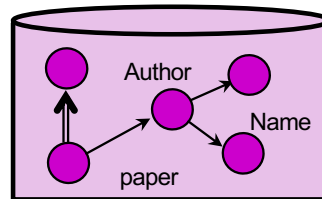
- Personalization: content adapted to user
 - upon system's decision
 - upon user's request
- Customization: structure adapted to user
 - according to the user's role
 - upon user's request
- System Dependent
 - Performance Reasons
- Context dependence
 - User, Device, Network, Place, Time, Rate



Web pages



RDBMS



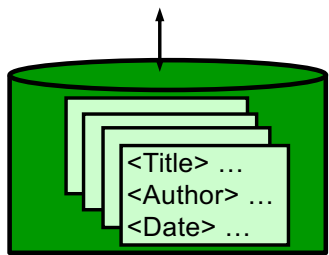
OODBMS



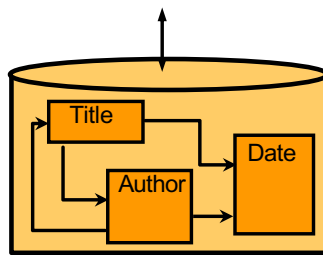


System level

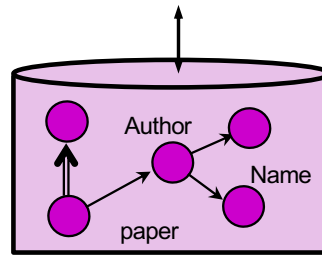
- Protocols
 - Predefined methods of communication



Web pages



RDBMS



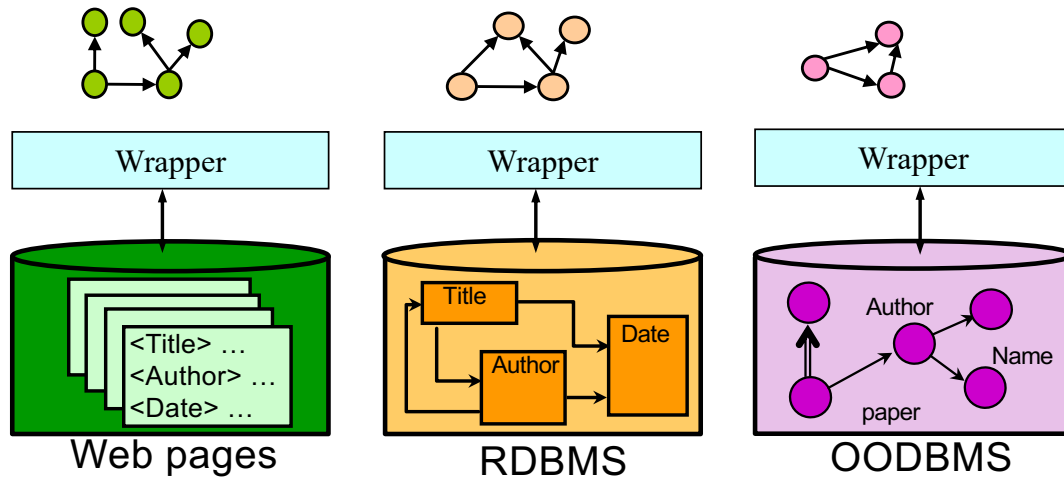
OODBMS



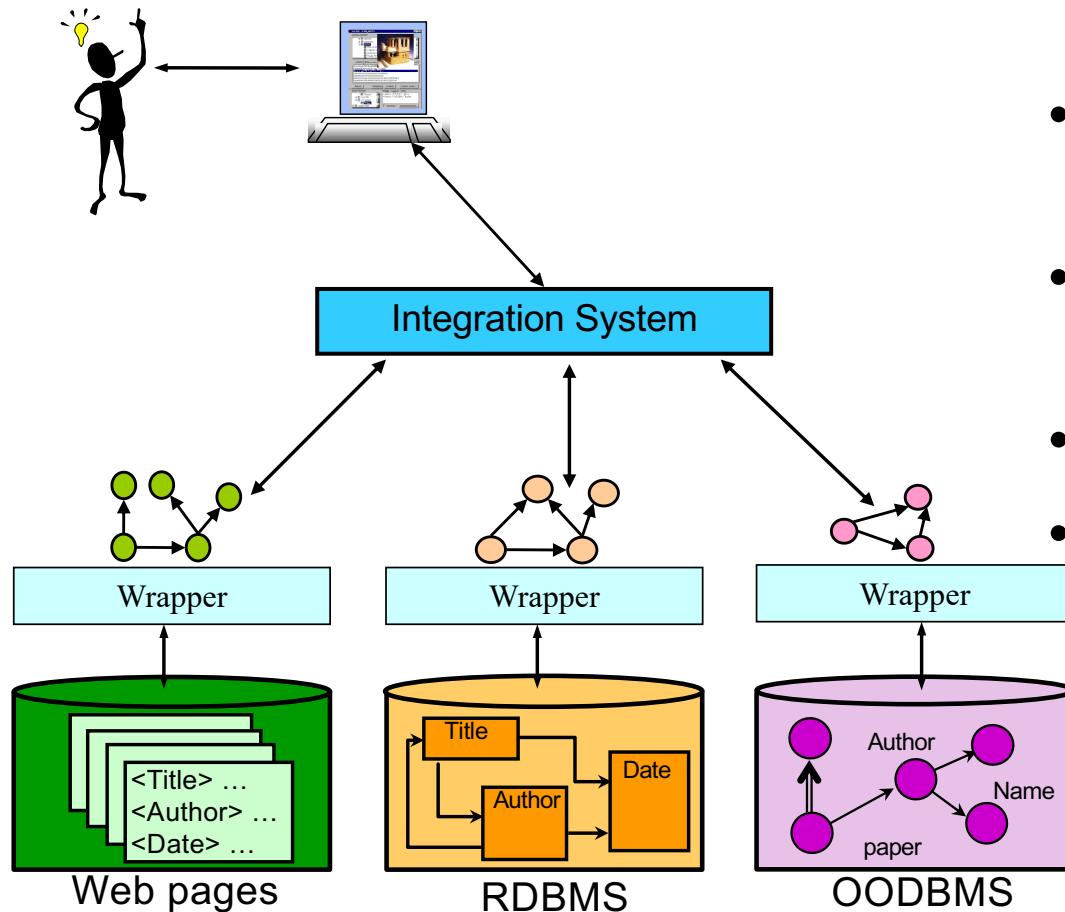


Model level

- Wrappers
 - Wrap the sources into a common data model



Structural and semantic level

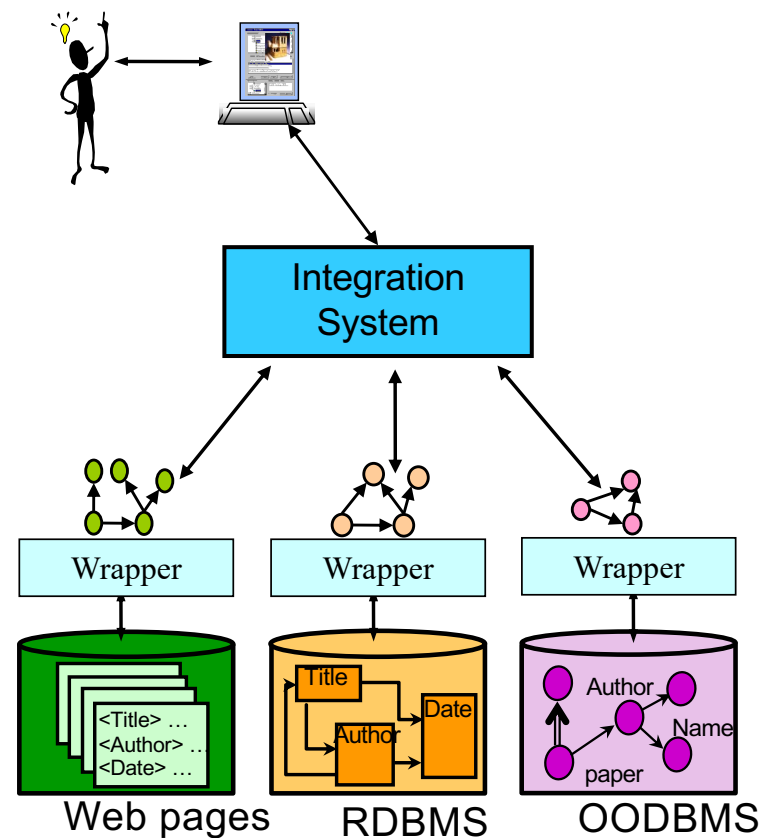


- Structural
 - Different structures for the same data
- Semantic
 - Different semantics of the same structure
- Integrated View
- Many architectures for this basic idea

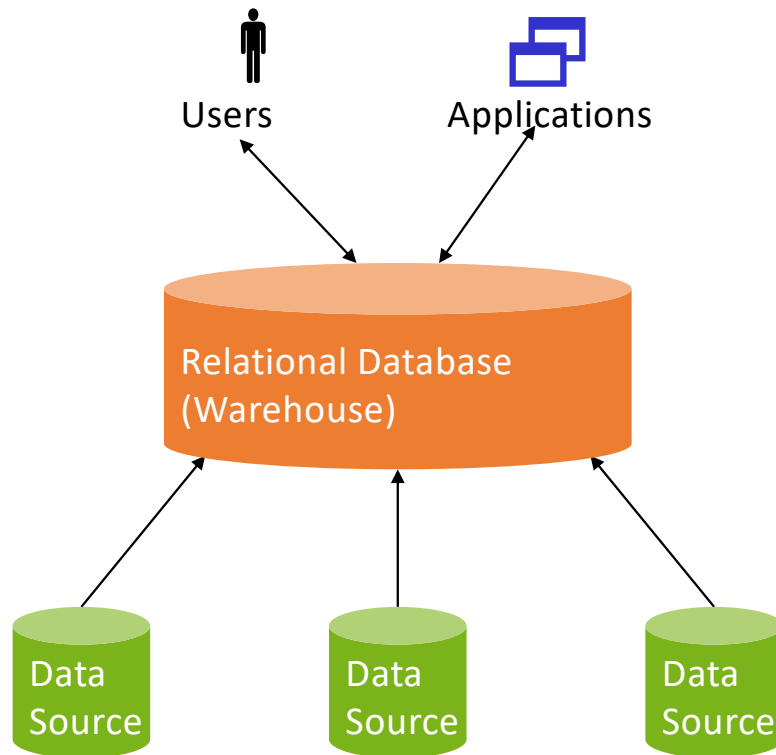


An information integration system

- A set of Local Databases
 - Each with a Local Schema & a Local Instance
- A Global Integrated Schema
- A set of Mappings
 - Between the Global and the Local Schemas
 - Describe the relationship between the data in the sources and the data as the user “sees” them



Data warehouse architecture



OLAP / Decision Support
Data Cubes / Data Mining

ETL Tools
(Extract-Transform-Load)

Data Cleaning



Virtual integration architecture

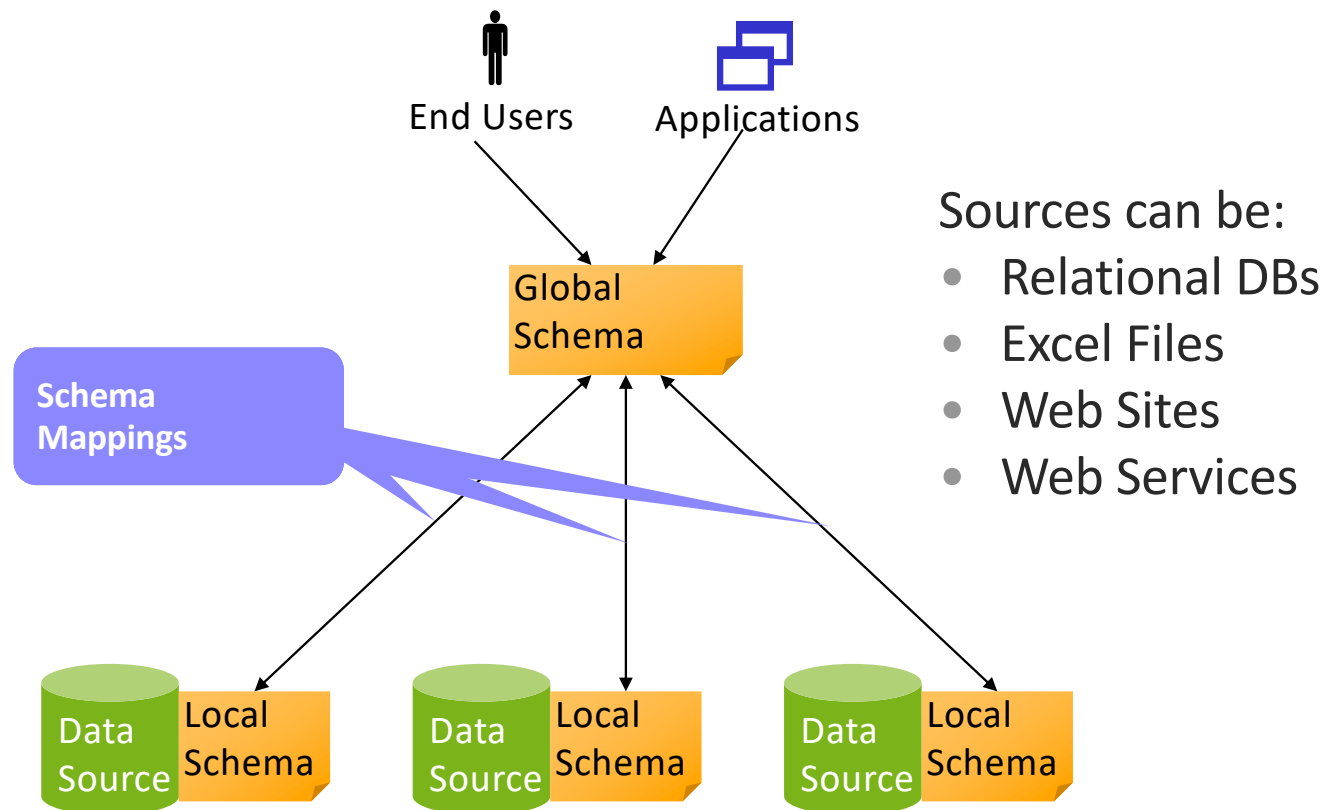
- Leave the data in the sources
- When a query comes in:
 - Determine the relevant sources to the query
 - Break down the query into sub-queries for the sources
 - Get the answers from the sources, filter them if needed and combine them appropriately
- Data is fresh
- Otherwise known as

On Demand Integration



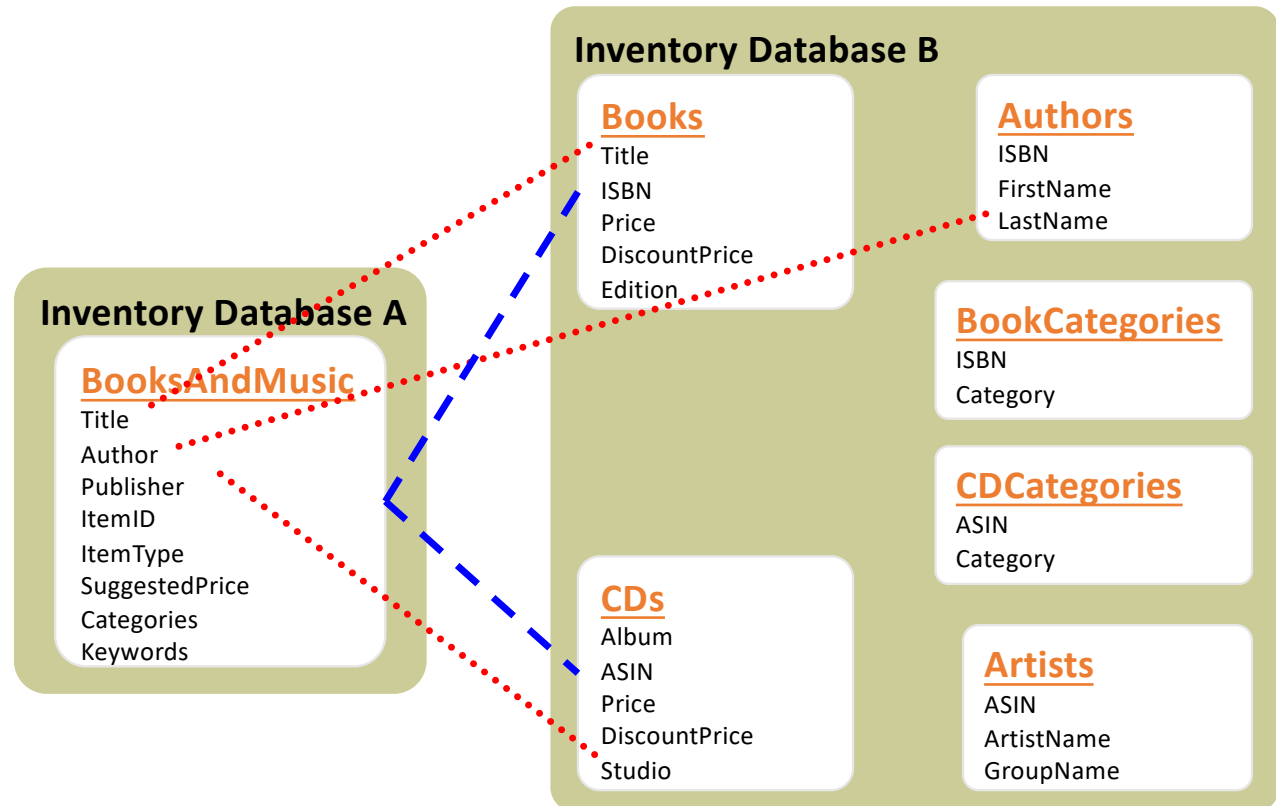
Virtual integration architecture

Design-Time



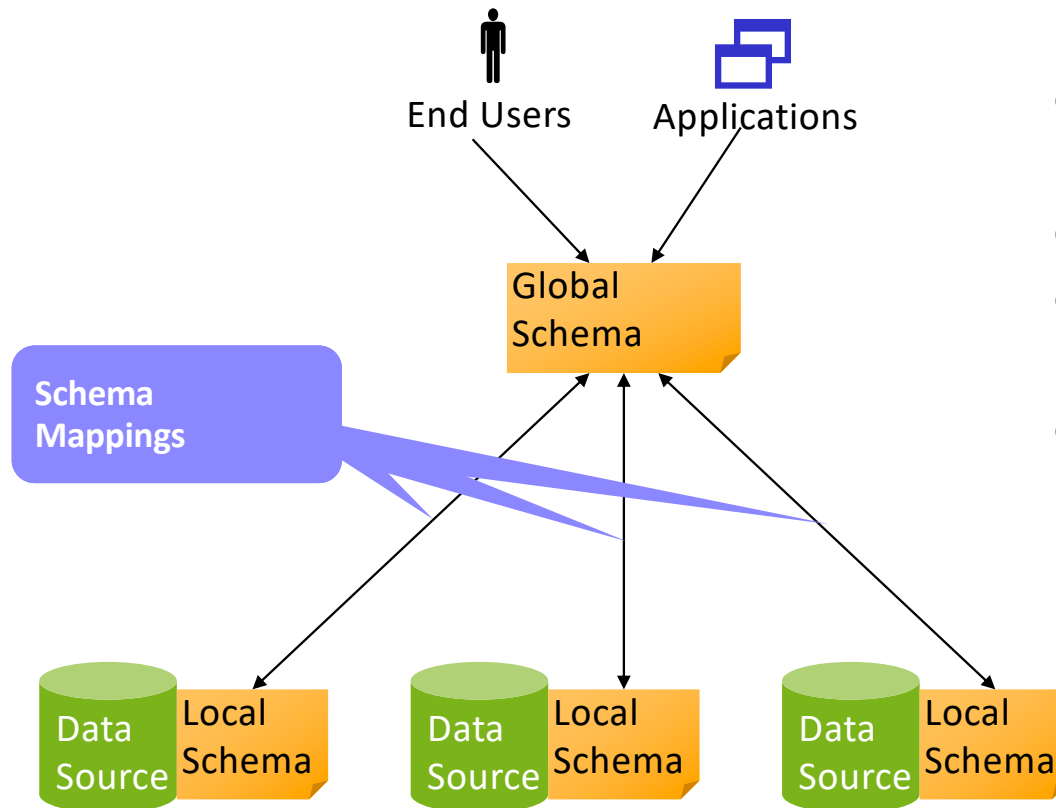
Schema mappings

- Differences in:
 - Names in schema
 - Attribute grouping
 - Coverage of databases
 - Granularity and format of attributes



Issues for schema mappings

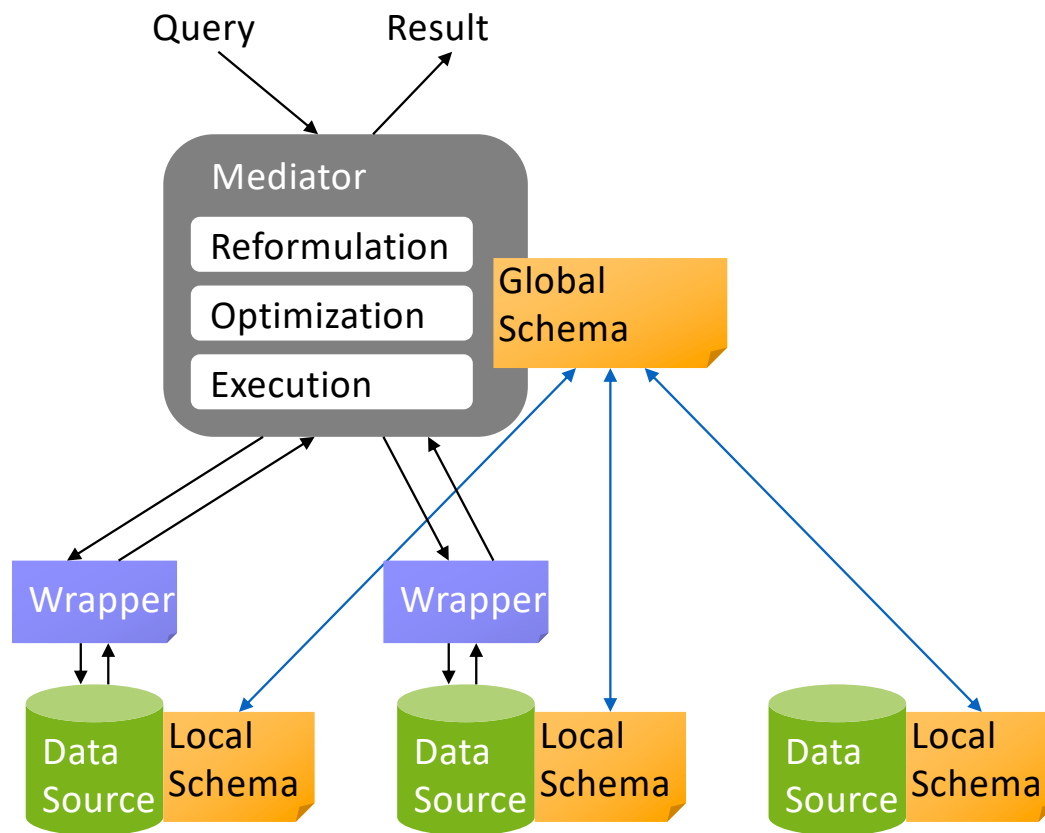
Design-Time



- What formalisms to express them?
- How to create them?
- Can we discover them somehow?
- How do we use them?

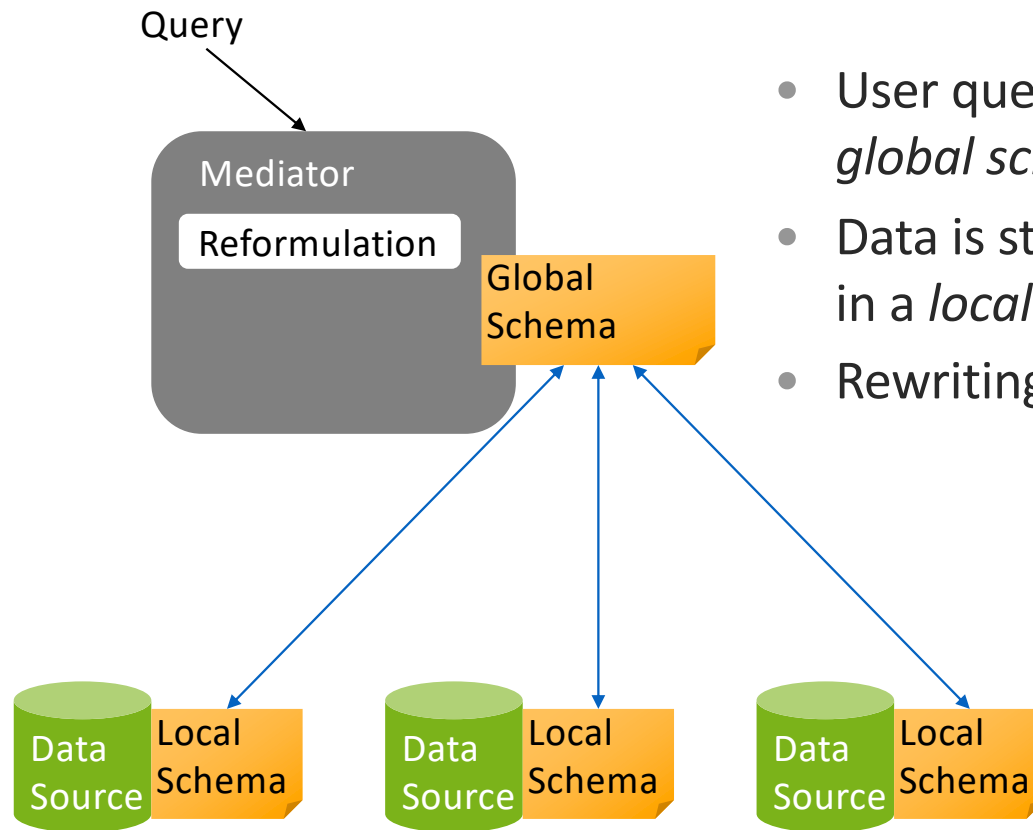
Virtual integration architecture

Run-Time



Issues for query processing

Reformulation



- User queries refer to the *global schema*
- Data is stored in the sources in a *local schema*
- Rewriting algorithms

Issues for query processing

Reformulation

Global Schema

Books

Title
ISBN
Price
DiscountPrice
Edition

```
SELECT ISBN, Price  
FROM Books  
WHERE Title = 'on the road'
```



Local Schema A

BooksAndMusic

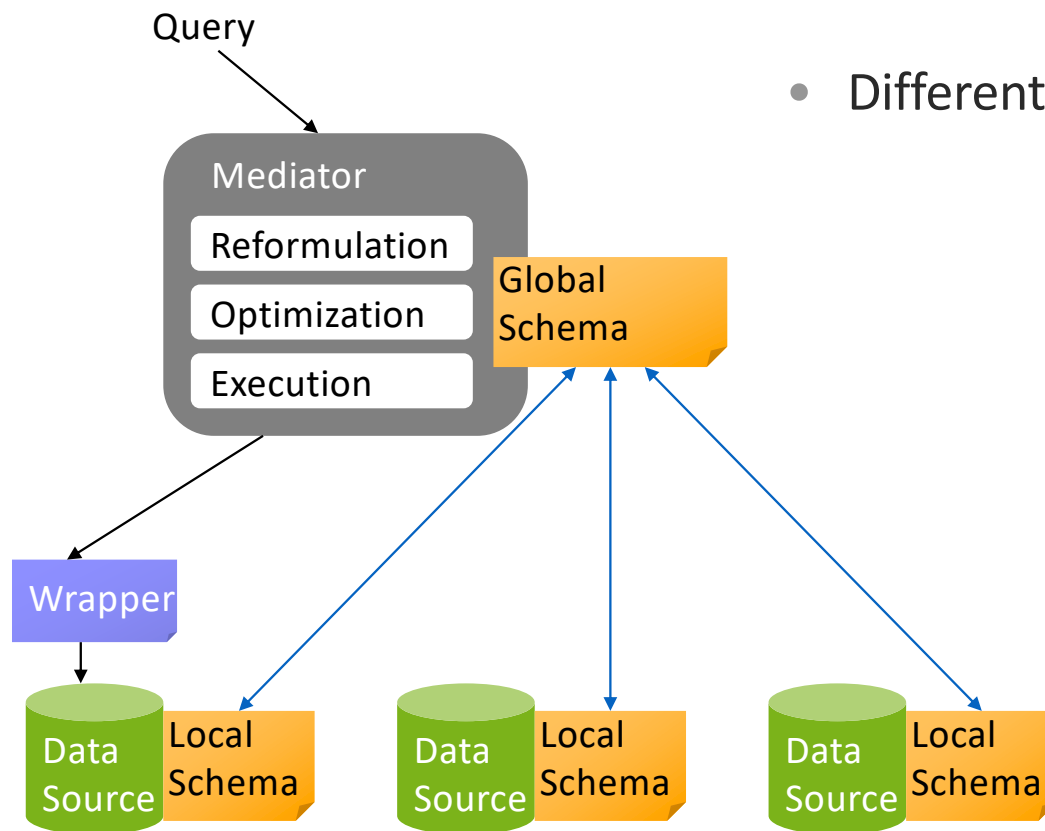
Title
Author
Publisher
ItemID
ItemType
SuggestedPrice
Categories
Keywords

```
SELECT ItemID, SuggestedPrice  
FROM BooksAndMusic  
WHERE Title = 'on the road'  
AND ItemType = 'Books'
```



Issues for query processing

Query Translation



- Different query languages



Issues for query processing

Query Translation

Global Schema

Books

Title
ISBN
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```
SELECT ISBN, Price  
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WHERE Title = 'on the road'
```

Local Source A

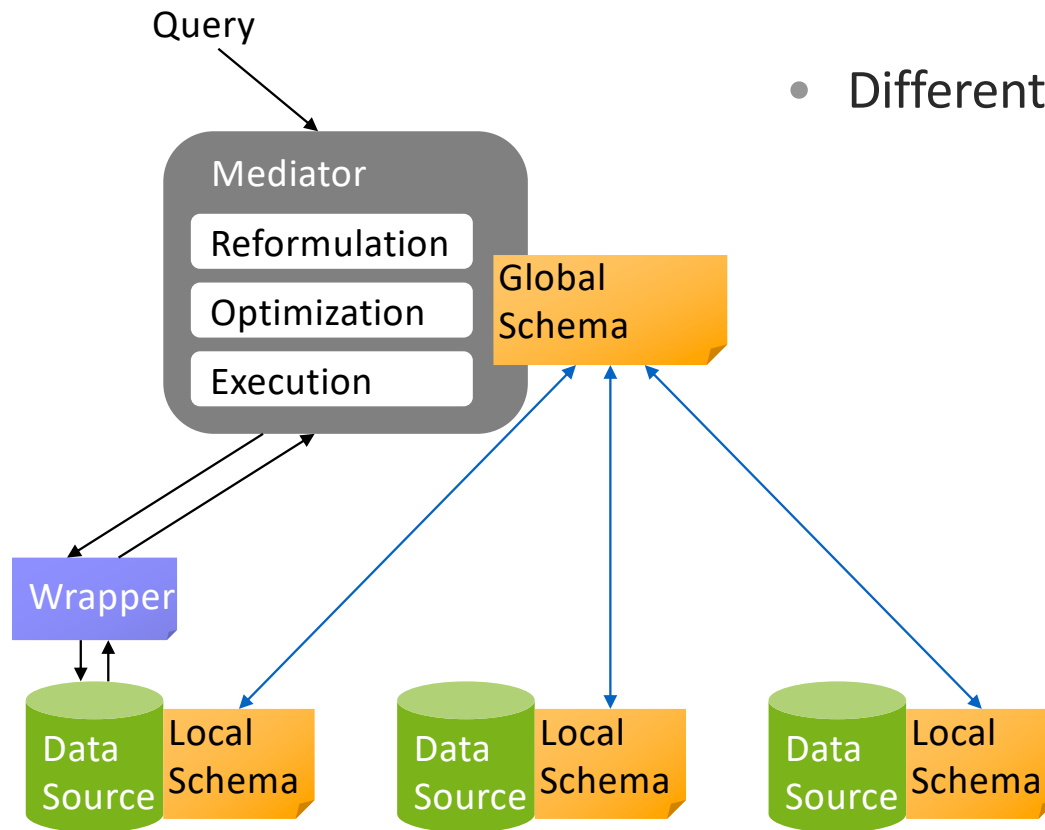


<http://www.amazon.com/homepage.html?ItemType=Books&Title=on+the+road>



Issues for query processing

Data Translation



- Different data models

Issues for query processing

Data Translation

Global Schema

Books

Title
ISBN
Price
DiscountPrice
Edition

Local Result A

 **On the Road** -- by Jack Kerouac; Paperback ([Rate it](#))
[Buy new](#): **\$10.86** -- [Used & new from](#): **\$5.90**

Title	ISBN	Price	...	...
On the Road	123	10.86	...	...

<table>
<tr>
 <td>

 On the Road

 -- by Jack Kerouac; Paperback

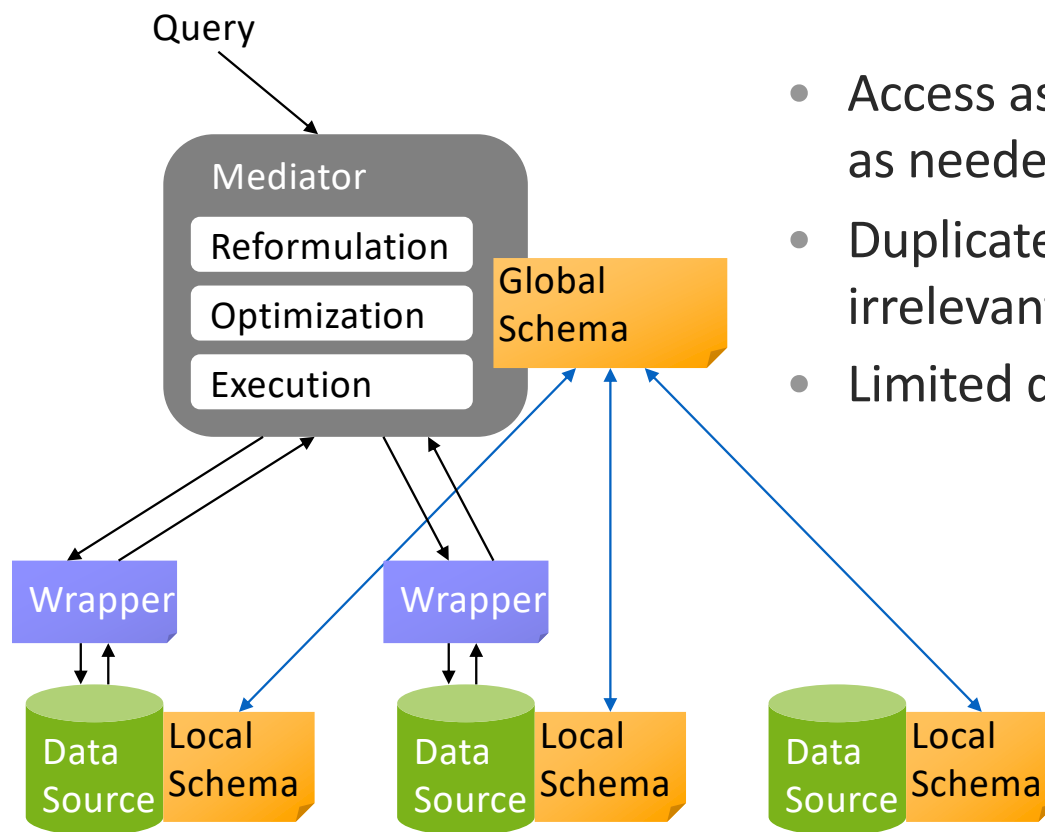
 Buy new

 :<b class=price>\$10.86
 </td>
</tr>
</table>



Issues for query processing

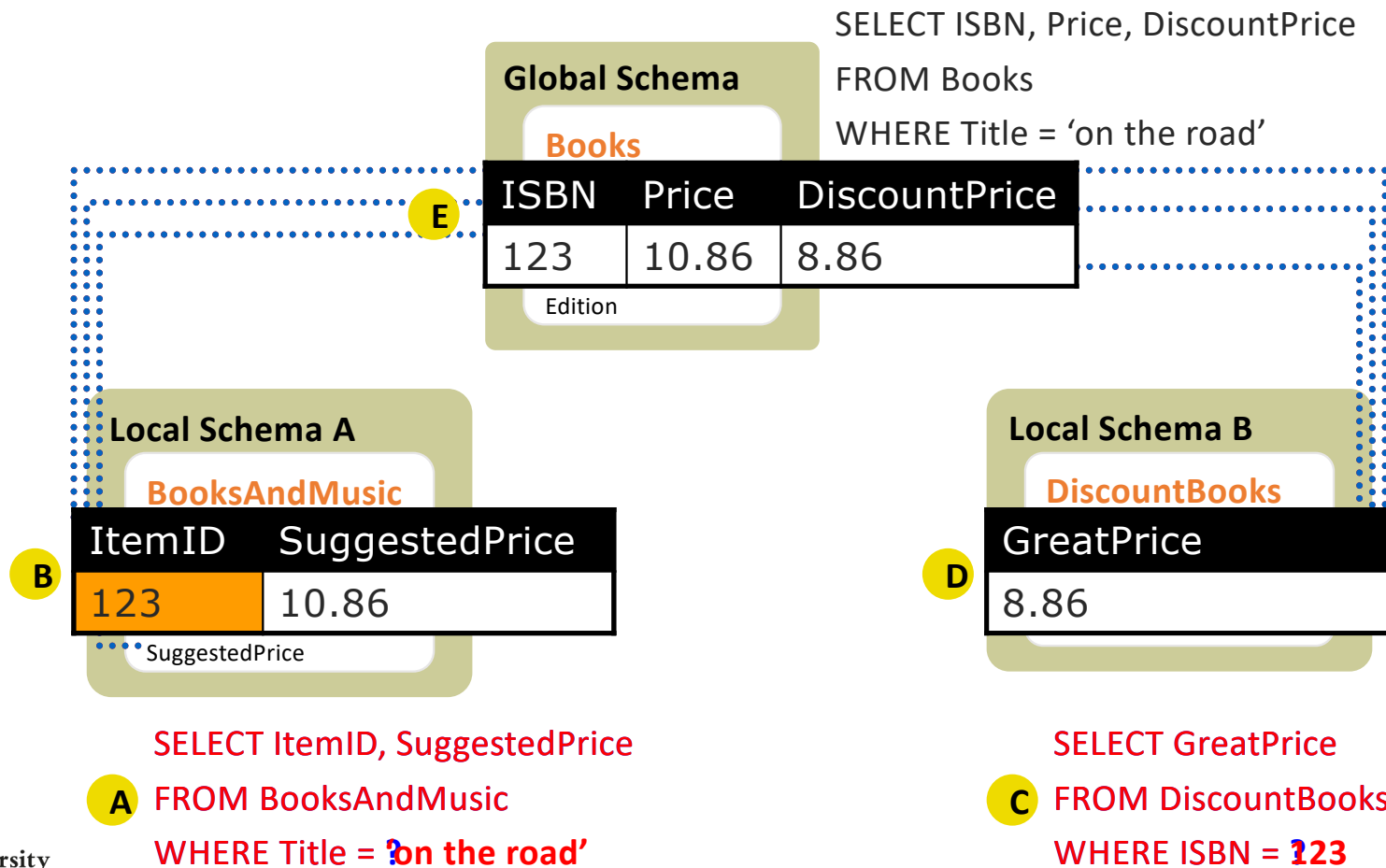
Query Execution



- Access as many data sources as needed
- Duplicate/redundant and irrelevant data
- Limited query capabilities

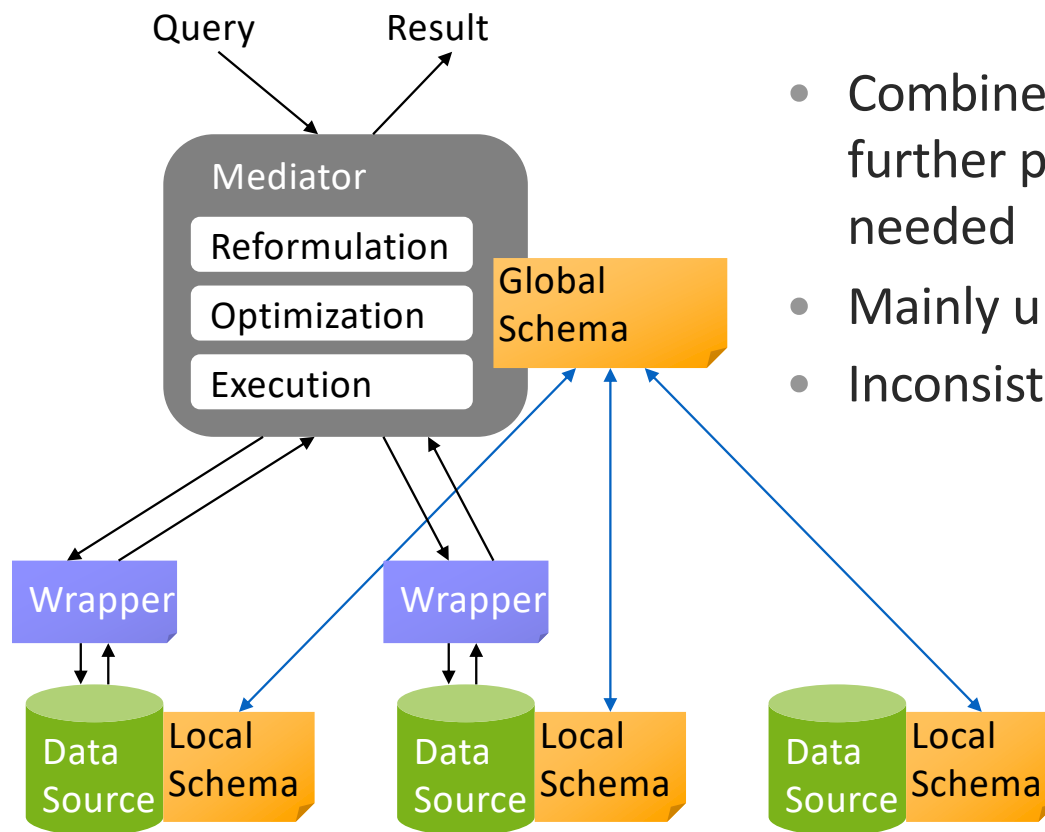
Issues for query processing

Limited Query Capabilities



Issues for query processing

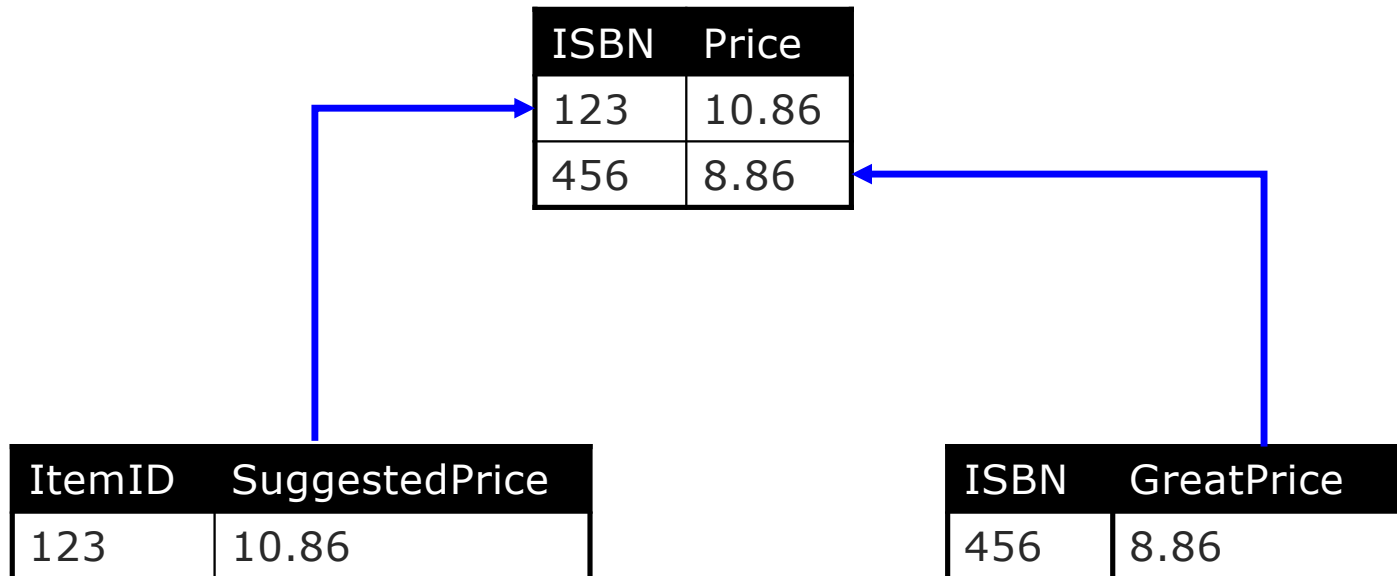
Query Answering



- Combine the results and further process them if needed
- Mainly union and merge
- Inconsistencies

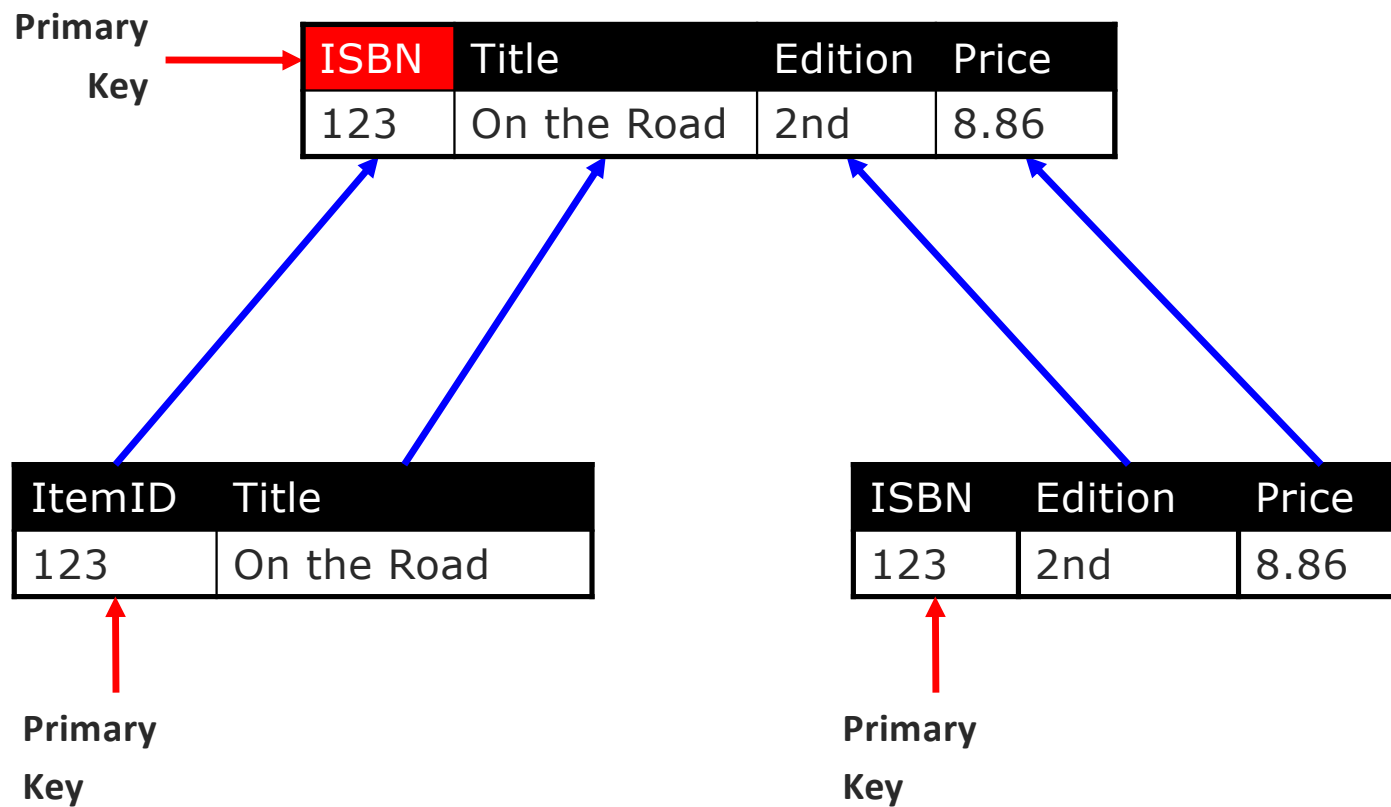
Issues for query processing

Query Answering (Union)



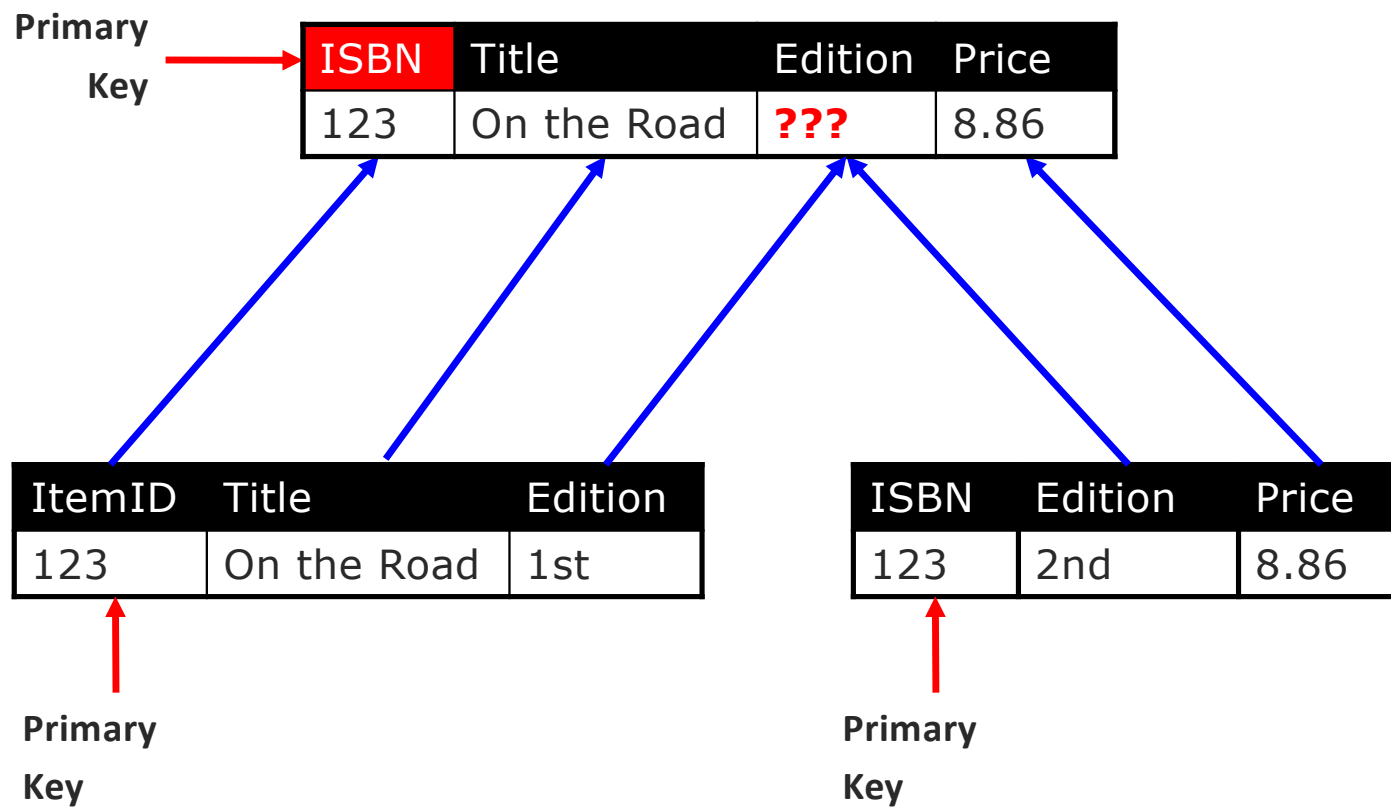
Issues for query processing

Query Answering (Merge)



Issues for query processing

Query Answering (Inconsistencies)



Mappings



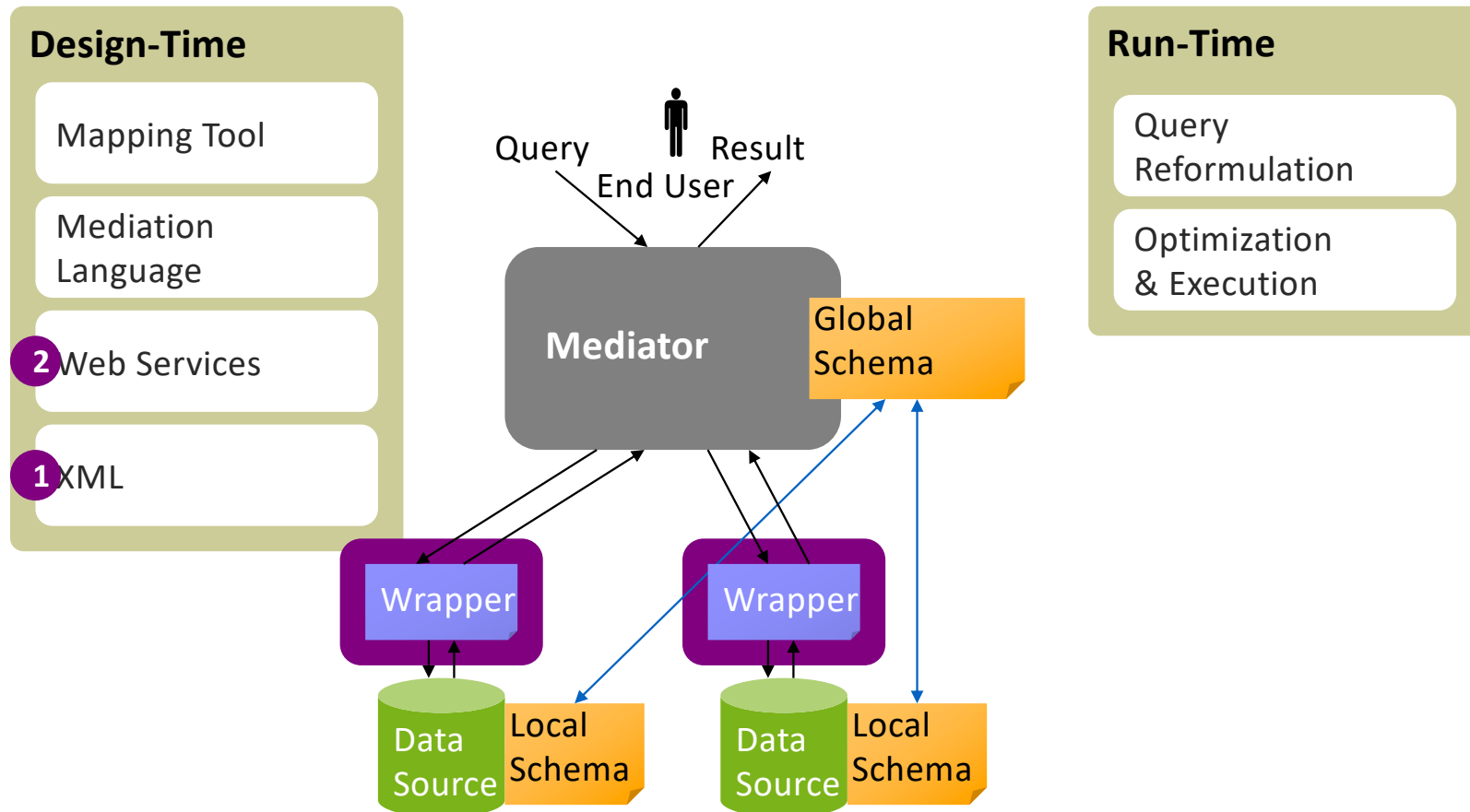
Virtual Integration Architecture

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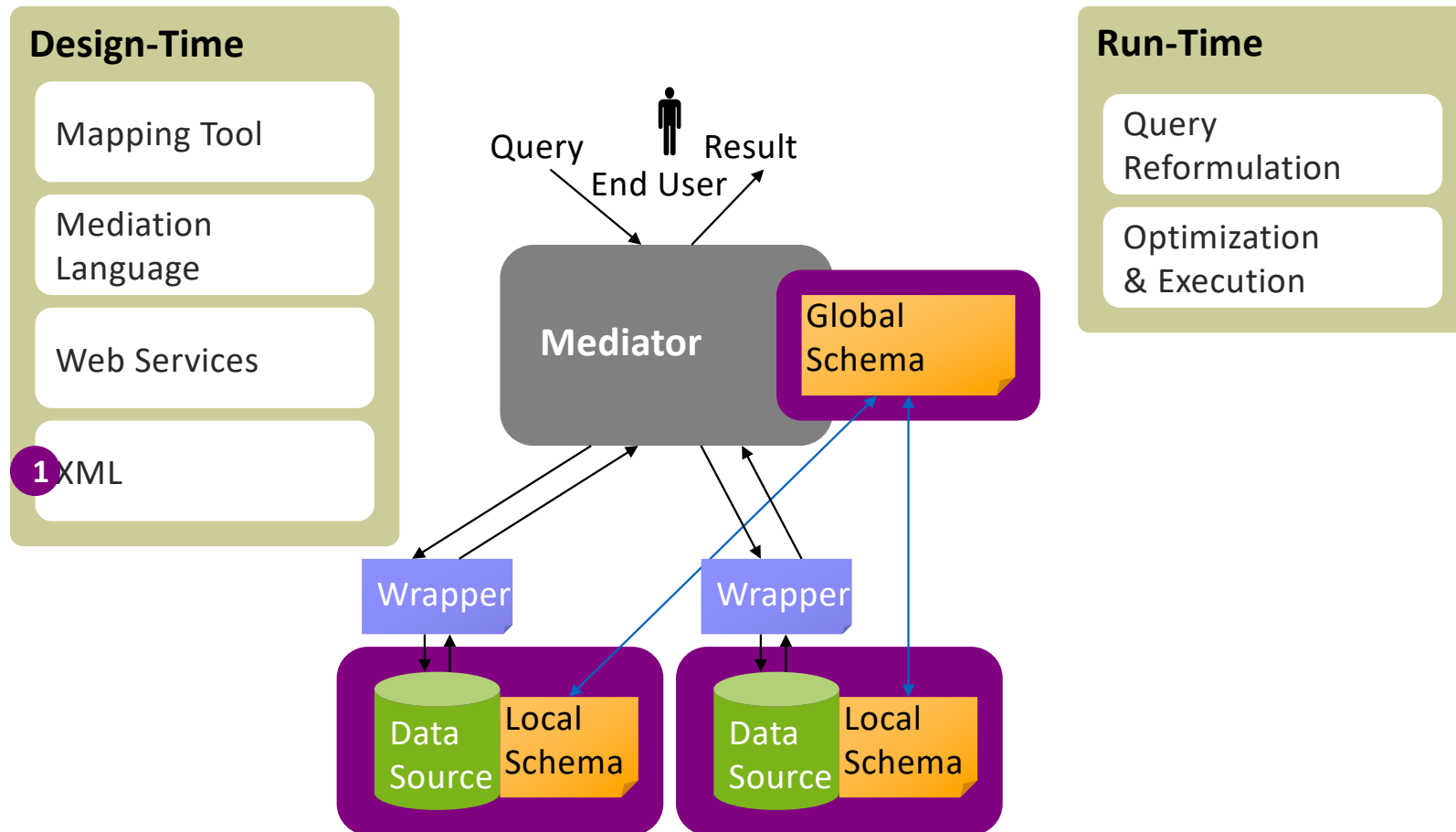
On Demand Integration



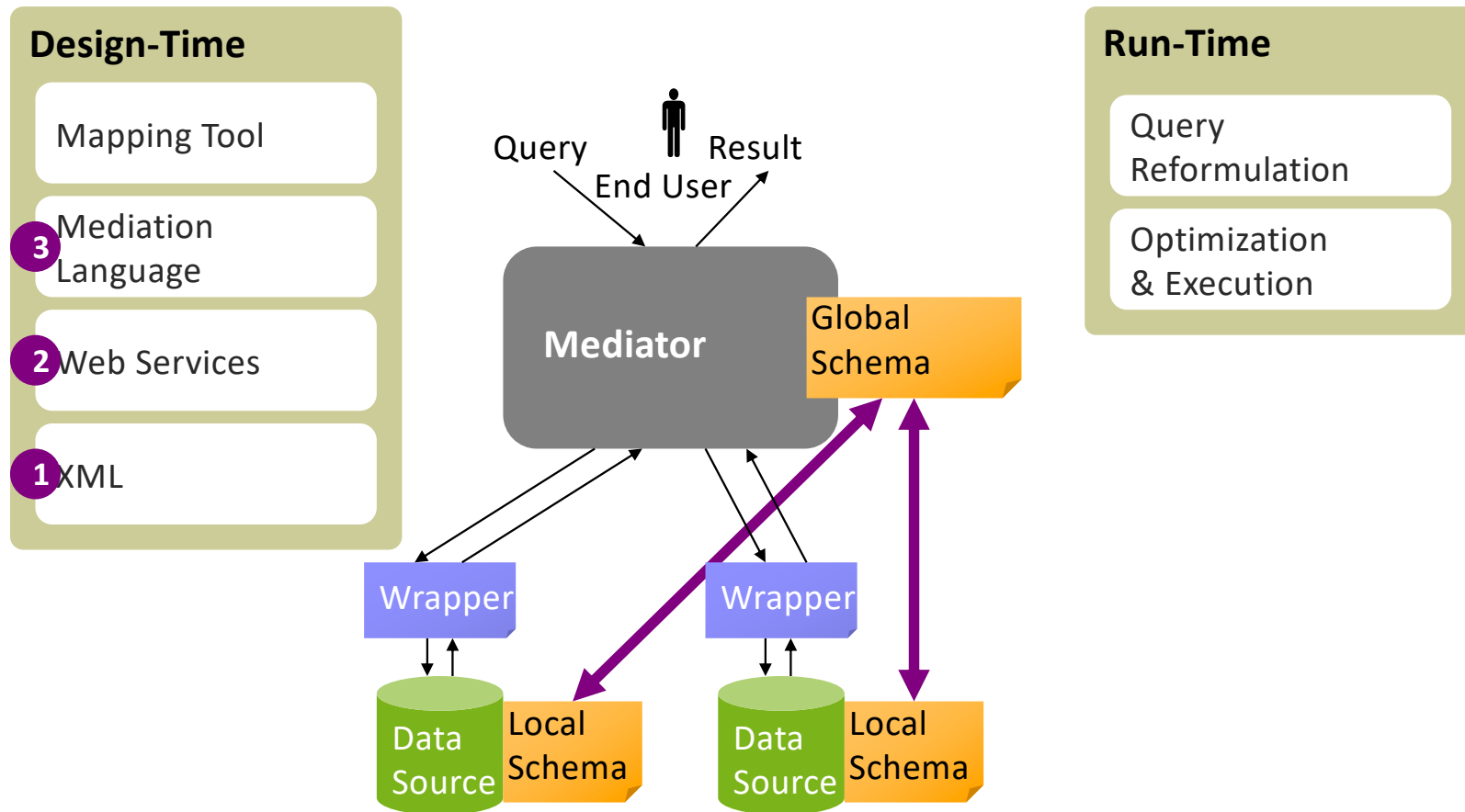
Virtual Integration Architecture



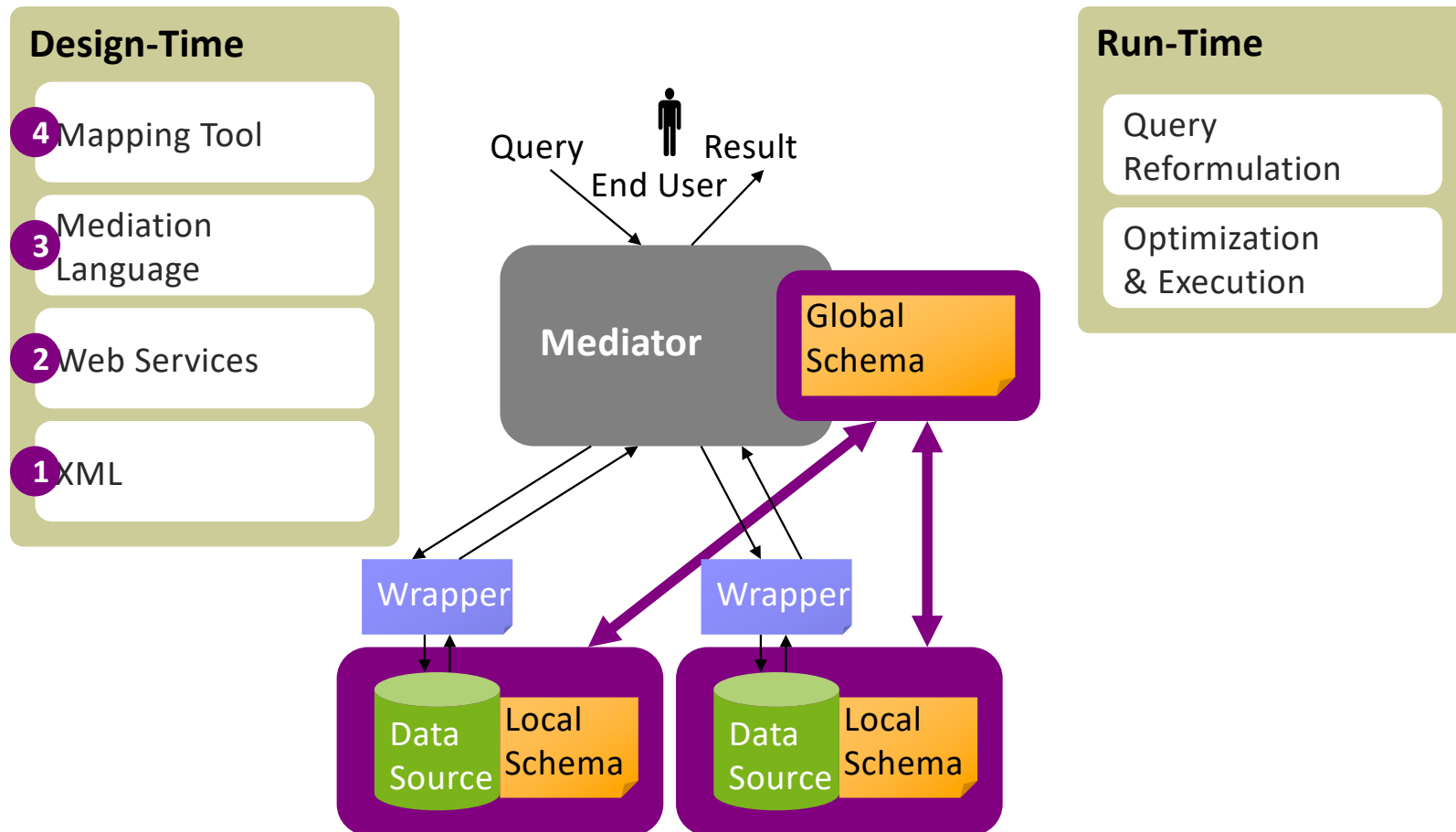
Virtual Integration Architecture



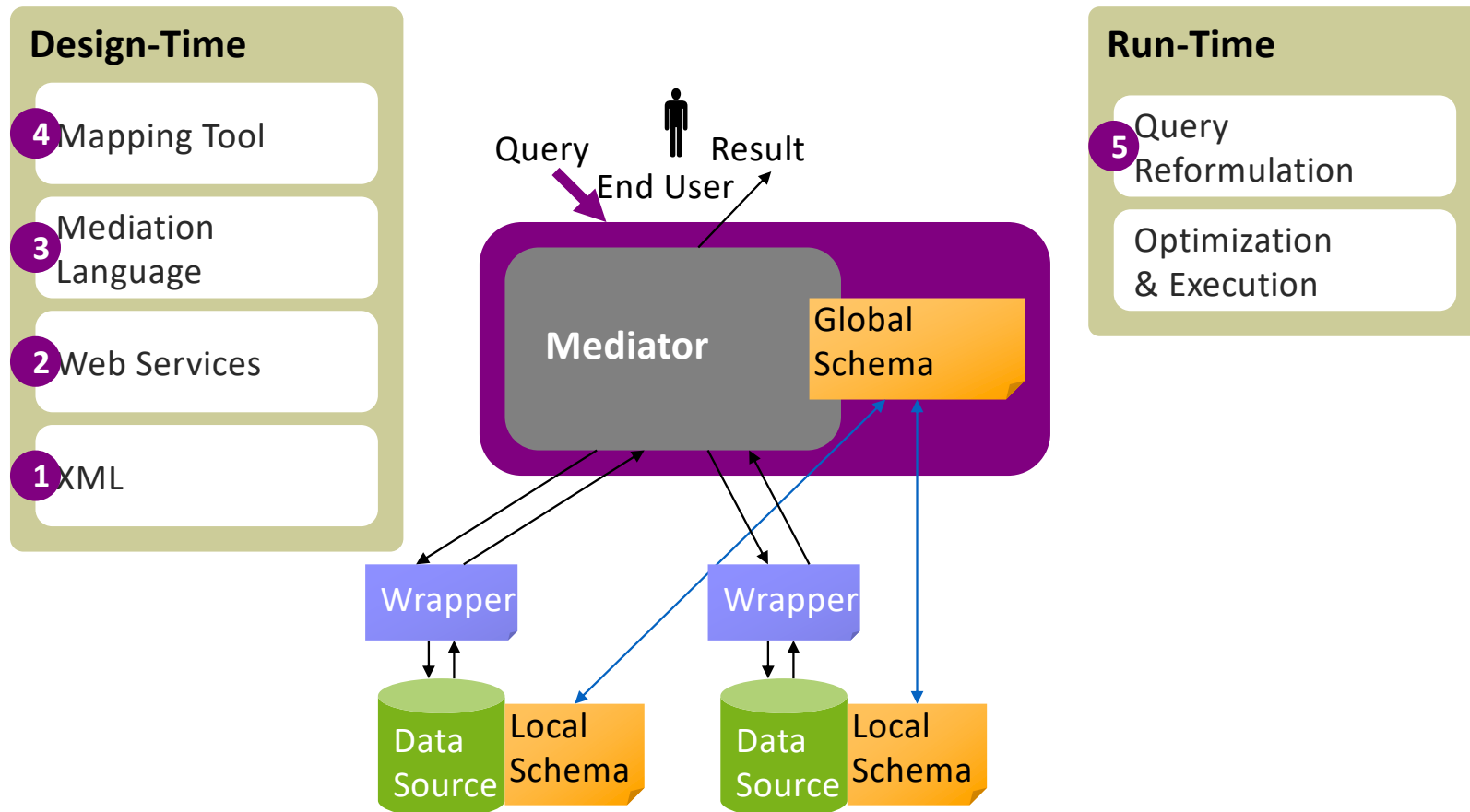
Virtual Integration Architecture



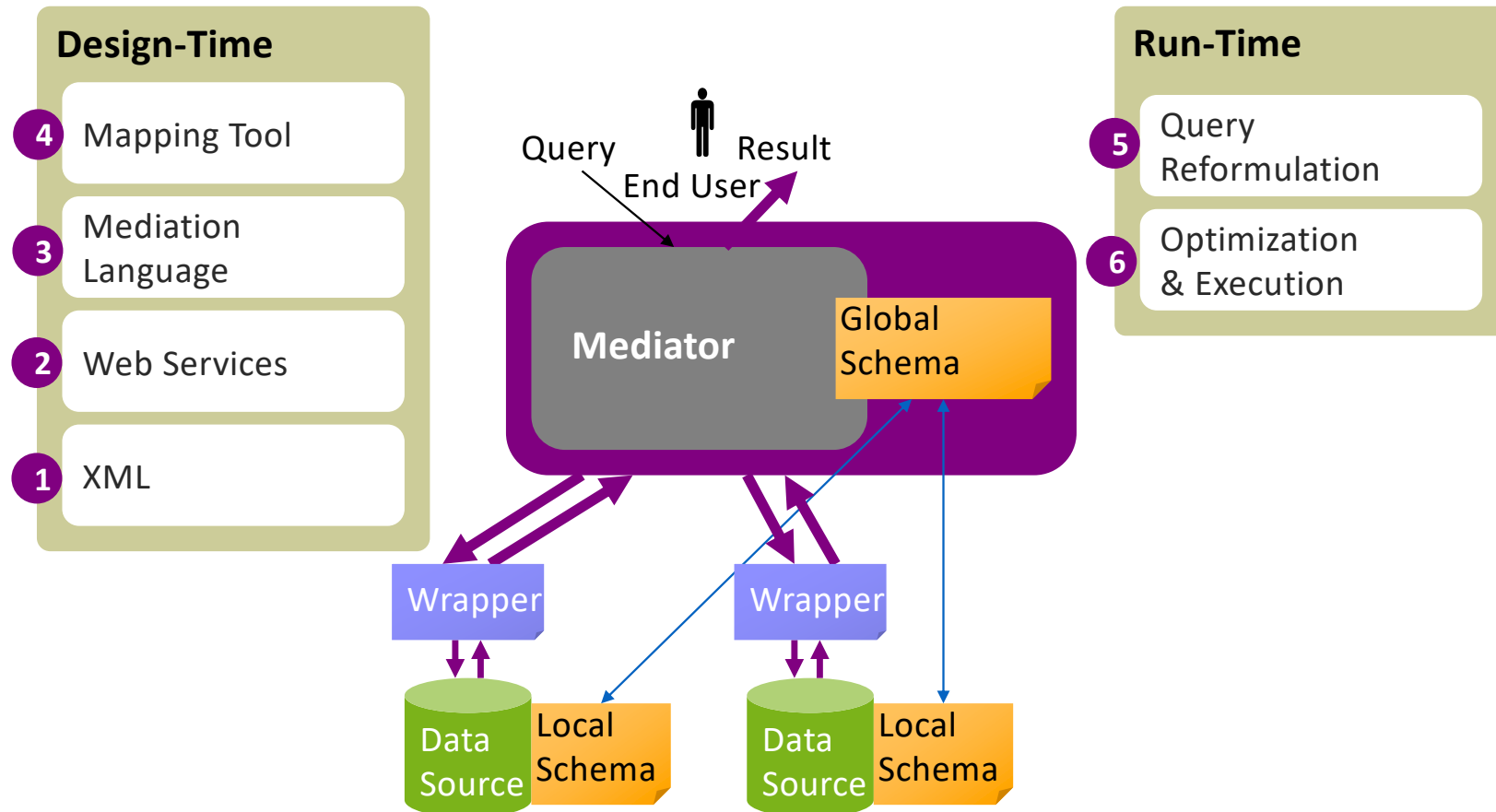
Virtual Integration Architecture



Virtual Integration Architecture



Virtual Integration Architecture



Virtual Integration Approaches

Dimensions to Consider:

- How many sources are we accessing?
- How autonomous are they?
- Meta-data about sources?
- Is the data structured?
- Queries or also updates?
- Requirements: accuracy, completeness, performance, handling inconsistencies.
- Closed world assumption vs. open world?

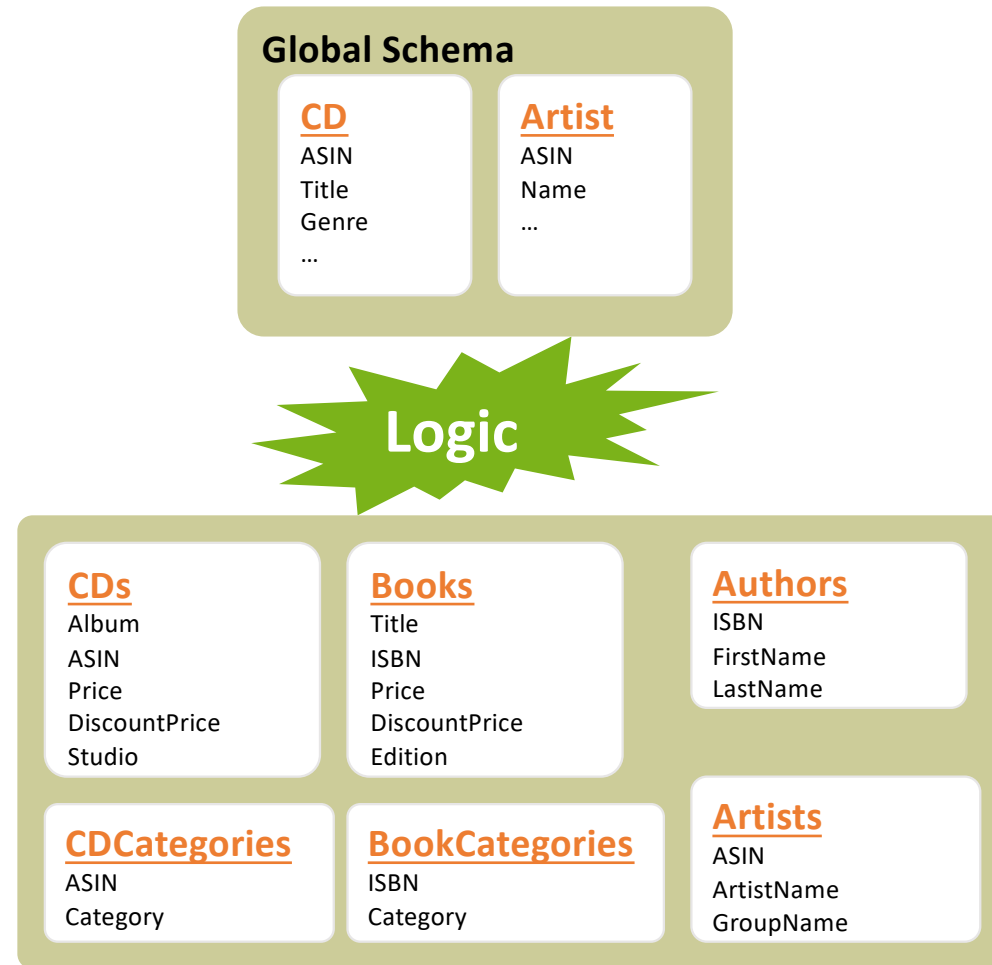


Requirements

- Relations and Attribute names
 - Names
- Tabular Organization
 - Splitting
- Granularity Level
 - More information
- Data Level Variations
 - Syntax
- Access Patterns Limitations
- Source completeness



Mediation Languages



Requirements from Source Descriptions

- **Expressive power:** distinguish between sources with closely related data. Hence, be able to prune access to irrelevant sources.
- **Easy addition:** make it easy to add new data sources.
- **Reformulation:** be able to reformulate a user query into a query on the sources efficiently and effectively.



Reformulation Problem

Given:

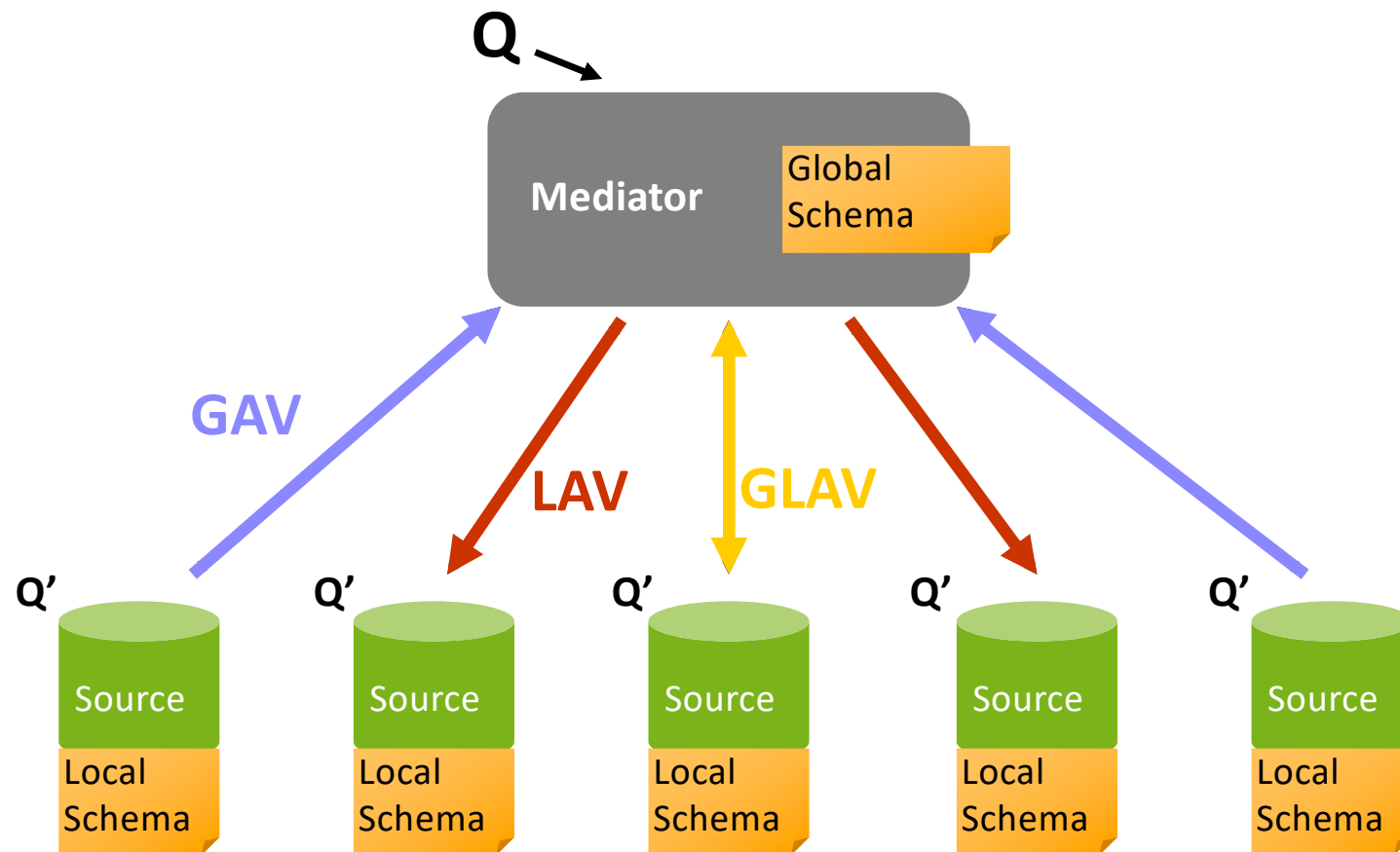
- A query Q posed over the global schema
- Descriptions of the data sources

Find:

- A query Q' over the data source relations, such that:
 - Q' provides only *correct answers* to Q , and
 - Q' provides *all* possible answers to Q given the sources.



Languages for Schema Mapping



Global-as-View (GAV)

Global Schema:

Movie(title, dir, year, genre)

Schedule(cinema, title, time)

Integrating View:

Create View Movie AS

```
SELECT * FROM S1
```

union

```
SELECT * FROM S2
```

union

```
SELECT S3.title, S3.dir, S4.year, S4.genre
```

```
FROM S3, S4
```

```
WHERE S3.title = S4.title
```

[S1(title, dir, year, genre)]



[S2(title, dir, year, genre)]



[S3(title, dir)]



[S4(title, year, genre)]



Global-as-View: Example 2

Global Schema:

Movie(title, dir, year, genre)

Schedule(cinema, title, time)

Integrating View:

```
Create View Movie(title, dir, year, genre) AS
  SELECT title, dir, year, NULL
  FROM S1
  union
  SELECT title, dir, NULL, genre
  FROM S2
```

[S1(title,dir,year)]



[S2(title,dir,genre)]



Global-as-View Summary

- Query reformulation boils down to view unfolding.
- Very easy conceptually.
- Can build hierarchies of global schemas.
- You sometimes lose information. Not always natural.
- Adding sources is hard. Need to consider all other sources that are available.



Local-as-View

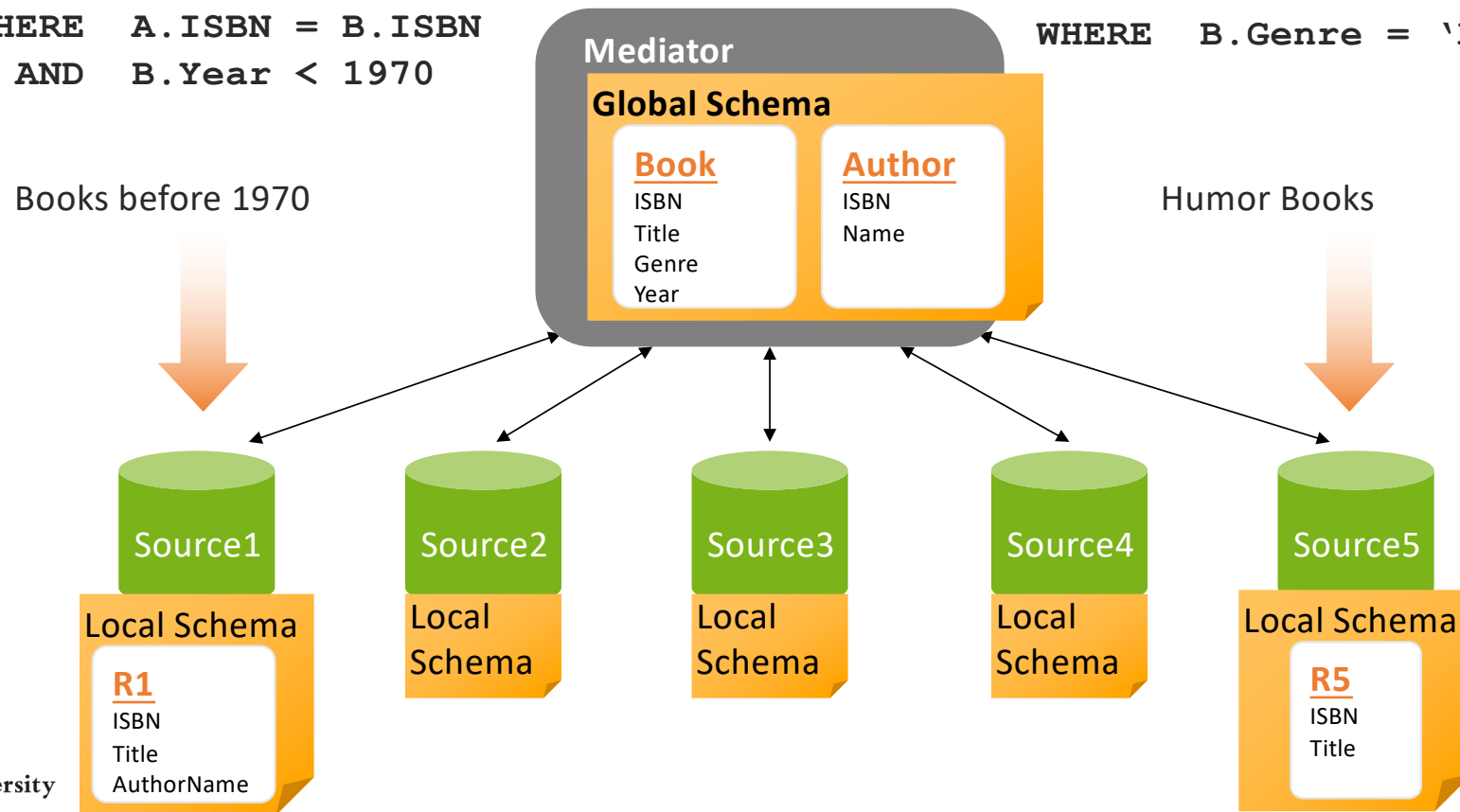
- Sources are defined in terms of the global schema using expression
- Every source provides expressions on how it can generate pieces of the global schema
- Mediator can combine these expressions to find all possible ways to answer a query



Local-as-View (LAV)

```
Create View R1 AS  
SELECT B.ISBN, B.Title, A.Name  
FROM   Book B, Author A  
WHERE  A.ISBN = B.ISBN  
       AND B.Year < 1970
```

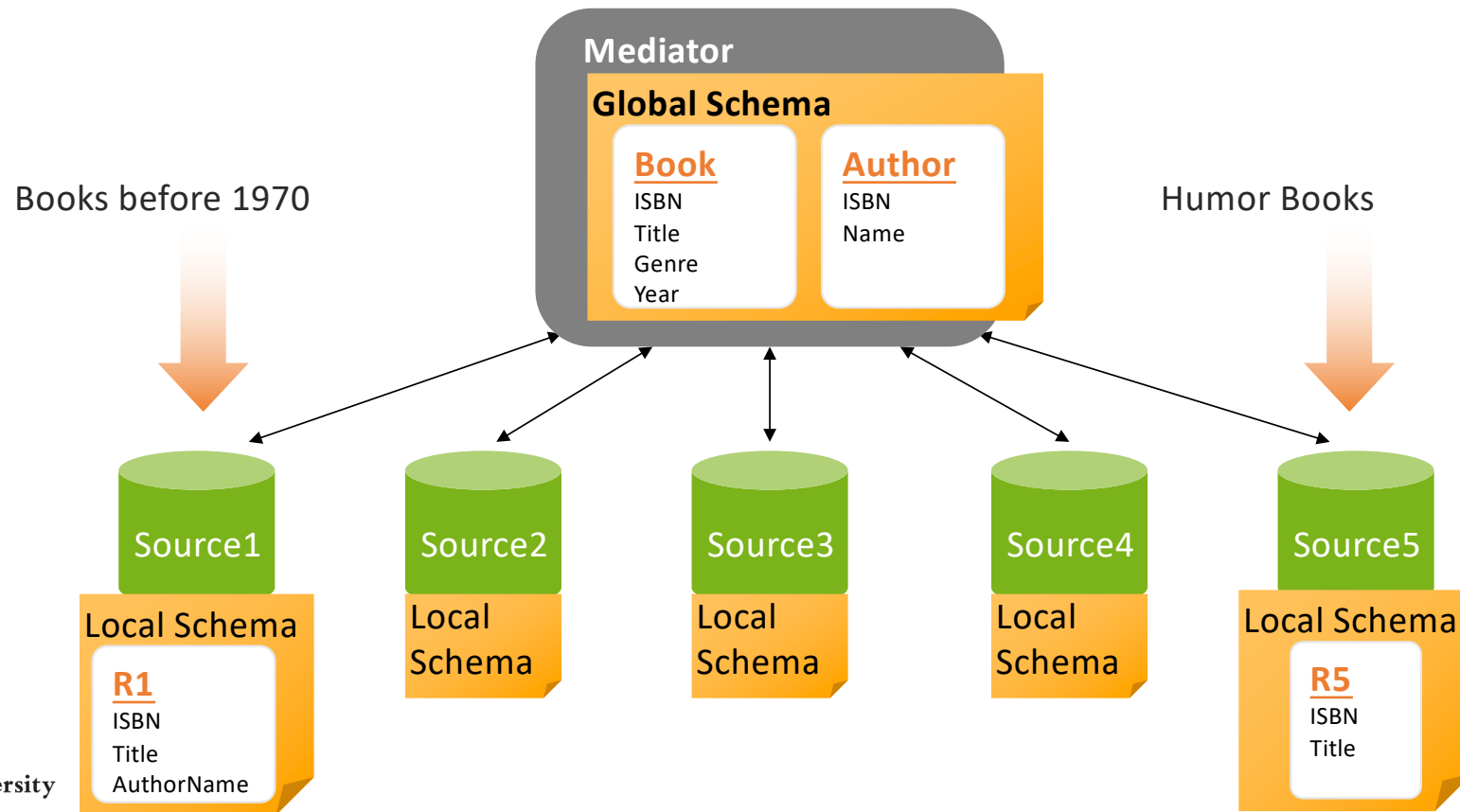
```
Create View R5 AS  
SELECT B.ISBN, B.Title  
FROM   Book B  
WHERE  B.Genre = 'Humor'
```



Query Reformulation

Query: Find authors of humor books

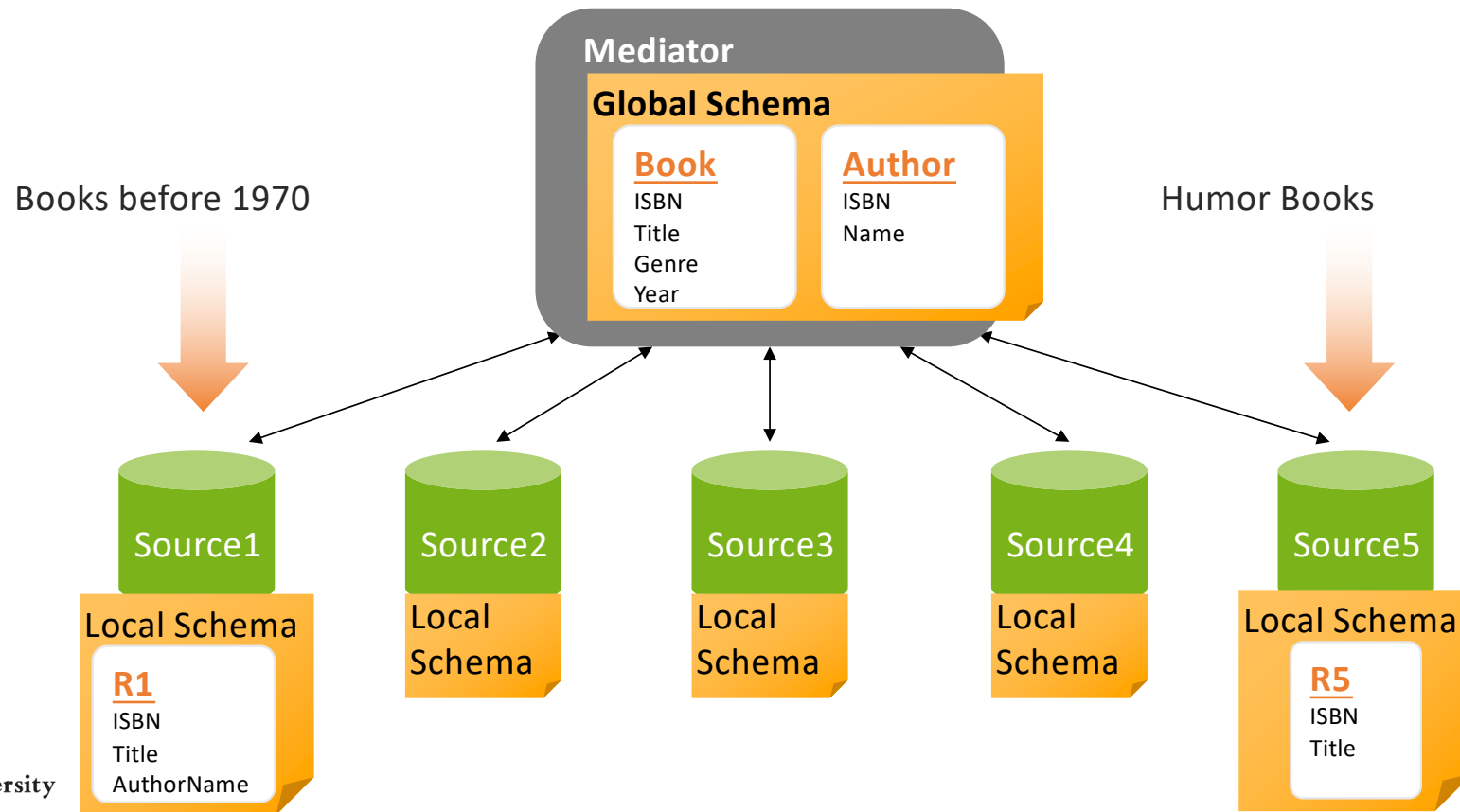
Plan: R1 Join R5



Query Reformulation

Query: Find authors of humor books before 1960

Plan: Can't do it!



Local-as-View: Example 1

Global Schema:

Movie(title, dir, year, genre)
Schedule(cinema, title, time)

Source Views:

```
Create Source S1 AS      [S1(title, dir, year, genre)]
      SELECT * FROM Movie
Create Source S3 AS      [S3(title, dir)]
      SELECT title, dir FROM Movie
Create Source S5 AS      [S5(title, dir, year)]
      SELECT title, dir, year
      FROM Movie
      WHERE year > 1960 AND genre='Comedy'
```



Local-as-View: Example 2

Global Schema:

Movie(title, dir, year, genre)

Schedule(cinema, title, time)

Source Views:

Create Source S4 [S4(cinema, genre)]

SELECT cinema, genre

FROM Movie M, Schedule S

WHERE M.title=S.title

*Now if we want to find
which cinemas are playing comedies, there is hope!*



Local-as-View Summary

- Very flexible. You have the power of the entire query language to define the contents of the source.
- Hence, can easily distinguish between contents of closely related sources.
- Adding sources is easy: they're independent of each other.
- Query reformulation: *answering queries using views!*



Additional References

- [Lenz02] Maurizio Lenzerini, Data integration: a theoretical perspective, PODS2002, <https://dl.acm.org/doi/10.1145/543613.543644>.
- [Mill18] Renee J. Miller, Open Data Integration, VLDB2018, <http://www.vldb.org/pvldb/vol11/p2130-miller.pdf>





- Summarize what you learned today in 2-minutes

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